

Soren L. Larsen

Chicago, IL

slarsen@berkeley.edu

Curriculum Vitae

ACADEMIC APPOINTMENTS

Postdoctoral Scholar, Department of Demography, University of California, Berkeley, May 2025 - present

Advised by Dr. Ayesha S. Mahmud.

EDUCATION

Doctor of Philosophy, Ecology, Evolution, and Conservation Biology, University of Illinois at Urbana Champaign, IL, Jan. 2021 - May 2025

Advised by Dr. Pamela P. Martinez (School of Molecular and Cellular Biology, Department of Statistics). Dissertation title: Host heterogeneity and respiratory disease dynamics: from structural inequity to immune history.

Master of Science, Mathematics, University of Illinois at Urbana Champaign, IL, Jan. 2021 - May 2022

Advised by Dr. Pamela P. Martinez.

Bachelor of Science, Mathematics, University of Washington, Seattle, WA, Sep. 2016 - Jun. 2020

Magna Cum Laude. Minors in Norwegian and the Comparative History of Ideas.

Complexity-GAINS International Summer School (Santa Fe Institute), Vienna, Austria, July 2022

Two week, NSF-funded workshop on disintegration of societies: quantitative modeling of complex socio-behavioral systems. Co-organized by the Santa-Fe Institute.

ADDITIONAL RESEARCH EXPERIENCE

Research Collaboration, COVID-19 Scenario Modeling Hub, May 2024 - present

- *Round 19*: Calibrated model to historical COVID-19 data and contributed projections for 2025-26. Co-wrote manuscript and conducted ensemble analysis with other modeling team leads.
- *Heterogeneity research round*: Mentored by Cecile Viboud (Fogarty International Center, NIH). Calibrated model and contributed projections for a Scenario Modeling Hub research round led by Samantha Bents (Stanford) on demographic heterogeneity in COVID-19 transmission and severity.

MARTIANS Student Intern, Sandia National Labs, May 2023 - Aug. 2023

- Worked with the epidemiology group led by Dr. Patrick Finley, exploring outcomes from socioeconomically-explicit vs. -agnostic models of SARS-CoV-2 transmission.

UNDERGRADUATE MENTEES

Alyssa Anastasi, Statistics, UIUC, 2025 - present || **Chloe Yang**, Statistics, UIUC, 2023 - present

Lydia Schieleit, Statistics, UIUC, 2024 || **Annika Venegas**, Mathematics, UIUC, 2023 - 2024

Erin Haslett, Political Science, UIUC, 2023 - 2024

Haylee West, Molecular and Cellular Biology, UIUC, 2021 - 2023

Jeffrin Joseph, Molecular and Cellular Biology, UIUC, 2021 - 2022 || **Ikgyu Shin**, Statistics, UIUC, 2022

Eliana Chandra, Statistics/Sociology, UIUC, 2022 || **Rafael Anorga**, Integrative Biology, UIUC, 2022

TEACHING EXPERIENCE

Guest Lecture, STAT 430, University of Illinois at Urbana-Champaign, March 5, 2026

- Lecture on serocatalytic modeling methods for infectious diseases, with an interactive lab in R.

Teaching Assistant, University of Illinois at Urbana-Champaign, Jan. 2021 - Dec. 2022

- Fall 2022: one Merit discussion section of Calculus 2 (Math 231). Wrote worksheets/mock exams and facilitated complex problem solving and collaboration.
- Spring 2022: two discussion sections of Calculus 2 (Math 231). Graded worksheets and helped develop exams.
- Fall 2021: two discussion sections of Calculus I (Math 221). Graded worksheets and exams.
- Spring 2021: two discussion sections of Calculus for Business (Math 234)

Math Grader, University of Washington, Winter/Summer Quarters 2019, Summer Quarter 2020

- Summer 2020: Graded homework for one section of Linear Analysis (MATH 309)
- Summer 2019/2020: Graded homework with feedback for one section of Introduction to Mathematical

Reasoning (MATH 300) each summer

- Winter 2019: Wrote homework keys and graded homework for two sections of Linear Analysis (MATH 309)

Volunteer Peer-Mentor, University of Washington, Spring/Fall 2018, Fall 2019

- Peer-mentors are part of the UW STEM-Dawgs program, whose purpose is to foster a sense of community within large undergraduate lecture courses, and address disparities in performance for women, underrepresented groups, and first-generation college students.
- Spring 2018: Trained in research-based pedagogical methods
- Fall 2018: Facilitated problem-solving and study skill development in a weekly discussion workshop for General Chemistry (CHEM 142), under the supervision of a graduate teaching assistant.
- Fall 2019: Facilitated an additional workshop with a graduate student

PEER-REVIEWED PUBLICATIONS (* indicates corresponding)

- **Larsen, S. L.***, Yang, J., Lv, H., Huan, Y. W., Teo, Q., Pholcharee, T., Lei, R., Gopal, A.B., Shao, E.K., Talmage, L., Mok, C.K.P., Takahashi, S., Kraay, A.N.M., Wu, N.C., & Martinez, P. P.* (2025). Reimagining the serocatalytic model for infectious diseases: a case study of common coronaviruses. *Epidemics*, 53, 100859.
- **Larsen, S.L.**, Noor, I., West, H., Chandra, E., Martinez, P. P., Kraay, A. N. M. (2025). Influence of immune history when choosing a SARS-CoV-2 booster strategy. *Scientific Reports*, 15(1), 35640.
- Kostandova, N., Schluth, C., Arambepola, R., Atuhaire, F., Bérubé, S., Chin, T., Cleary, E., Cortes-Azuero, O., Garcia-Carreras, B., Grantz, K.H., Hitchings, M.D.T., Huang, A.T., Kishore, N., Lai, S., **Larsen, S.L.**, Loiseate, S., Martinez, P., Meredith, H.R., Purbey, R., Ramiadantsoa, T., Read, J., Rice, B.L., Rosman, L., Ruktanonchai, N., Salje, H., Schaber, K.L., Tatem, A.J., Wang, J., Cummings, D.A.T., Wesolowski, A. (2024). A systematic review of using population-level human mobility data to understand SARS-CoV-2 transmission. *Nature Communications*, 15(1), 1-12.
- **Larsen, S. L.**, & Kraay, A. N. (2024). Transparent transmission models for informing public health policy: the role of trust and generalizability. *Proceedings of the Royal Society B*, 291(2023), 20232273.
- **Larsen, S. L.**, Shin, I., Joseph, J., West, H., Anorga, R., Mena, G. E., Mahmud, A.S., & Martinez, P. P. (2023). Quantifying the impact of SARS-CoV-2 temporal vaccination trends and disparities on disease control. *Science Advances*, 9(31), eadh9920.

PREPRINT/UNDER REVIEW(* indicates corresponding, ^ indicates shared first author)

- **Larsen, S. L.***, ^, Yang, J.^, Haslett, E. M., Anastasi, A., Venegas, A., Schieleit, L., Mahmud, Ayesha S., & Martinez, P. P.* (2026). Same household, different choices: variation in health behaviors related to respiratory viruses in Illinois. *medRxiv*, 2026-05.
- **Larsen, S. L.***, Martinez, P. P., & Mahmud, A. (2026). Changing COVID-19 vaccine eligibility could reshape disease burden for all. *medRxiv*, 2026-04.
- Bents, S.J., **Larsen, S.L.**, Jung, S., Howerton, E., Contamin, L., Loo, S.L., Gozzi, N., Davis, J.T., Vespignani, A., Chinazzi, M., Perra, N., Bandekar, S.R., Bi, K., Bouchnita, A., Fox, S.J., Meyers, L.A., Srivastava, A., Al Aawar, M., Chen, J., Venkatramanan, S., Hoops, S., Lewis, B., Bhattacharya, P., Machi, D., Marathe, M., Pei, S., Yao, Q., Yang, Q., Hill, A., Lemaitre, J., Nande, A., Willard, T., Pearson, C.A.B., Przykucki, E., McKee, C., Macdonald, J., Alleman, T., Smith, C.P., Carcelen, E., Yan, K., Hochheiser, H., Viboud, C., Lessler, J., Shea, K., León, T.M., Truelove, S. (*Under review*) Multi-model insights into racial and ethnic heterogeneities in early COVID-19 transmission and severity in the U.S.
- Liu, H., Yang, J., **Larsen, S. L.**, Martinez, P. P., & Dayanikli, G. (2025). Incorporating Authority Perception, Economic Status, and Behavioral Response in Infectious Disease Control. *arXiv preprint arXiv:2512.23188*.
- Nande, A., **Larsen, S.L.**, Turtle, J., Davis, J.T., Ramdas Bandekar, S., Lewis, B., Chen, S., Contamin, L., Jung, S., Howerton, E., Shea, K., Bay, C., Ben-Nun, M., Bi, K., Bouchnita, A., Chen, J., Chinazzi, M., Fox, S.J., Hill, A.L., Hochheiser, H., Lemaitre, J.C., Loo, S.L., Marathe, M., Meyers, L.A., Pearson, C.A.B., Porebski, P., Przykucki, E., Smith, C.P., Venkatramanan, S., Vespignani, A., Willard, T.C., Yan, K., Viboud, C., Lessler, J., Truelove, S. (2026) Assessing the Impact of Timing and Coverage of United States COVID-19 Vaccination Campaigns: A Multi-Model Approach. *medRxiv preprint 2026.04.07.26349269*.

OTHER PUBLICATIONS

- **Larsen, S.L.**, Noor, I., West, H., Chandra, E., Martinez, P.P., Kraay, A.N.M. "Disentangling the impact of

natural infections and vaccination in a rapidly evolving pandemic: prioritizing limited vaccine supplies in low- and middle-income countries" WHO pink book. Meeting of the Advisory Committee on Immunization and Vaccines related Implementation Research. February 2023, pages 215-271.

- **Larsen, S. L.,** Beyeler, W. E., Acquesta, E. C. S., Klise, K. A., & Finley, P. D. (2023). *Socioeconomically-inspired modeling to justify use of fine-grain mobility data* (No. SAND2023-07730). Sandia National Lab.(SNL-NM), Albuquerque, NM (United States).

GRANTS

SHIELD Illinois, 2023-2025 (Co-Investigator), \$100,000

Disentangling the effect of racial, ethnic, and socioeconomic health disparities on respiratory disease transmission in Illinois before and during the COVID-19 pandemic

HONORS AND AWARDS

Isabel Norton Award, Spring 2024

PEEC Travel Grant, Spring 2024

Best Talk, Annual Microbiology Research Conference, Monticello, Illinois, October 2023

List of Teachers Ranked as Excellent, Spring 2021 - Fall 2022

Barbara Sando Scholarship in Mathematics, awarded for Winter/Spring 2020

Phi Beta Kappa, inducted Spring 2019

Annual Dean's List, University of Washington, 2016-17 and 2017-18

TALKS AT MEETINGS, CONFERENCES AND SEMINARS

Mathematical and Statistical Modeling of Spread and Mitigation of Epidemics, Institute of Mathematical and Statistical Innovation, University of Chicago, May 2026

Invited participant - 25 minute research talk.

Inequalities in Infectious Disease Dynamics Symposium, London School of Hygiene and Tropical Medicine, April 2026

Sociodemographic disparities in health seeking behavior for respiratory viruses - 7 minute research talk at a public symposium. Additionally participated in a closed two-day workshop on social inequalities and infectious disease modeling.

Berkeley Population Biology of Infectious Diseases Seminar Series, February 2026

Understanding the role of sociodemographic disparities in shaping respiratory disease dynamics - 25 minute research talk

Berkeley Demography Brownbag Series, February 2026

Understanding the role of social disparities in shaping respiratory disease dynamics - 40 minute research talk

Scenario Modeling Hub Annual Meeting, September 2025

Presentation of Round 19 results with co-authors - 15 minute research talk

Epidemiology and Population Studies modeling seminar, NIH Fogarty International Center, Bethesda, Maryland, February 2025

Reimagining the serocatalytic model for infectious diseases: a case study of common coronaviruses - 50-minute research talk

Models of Infectious Disease Agent Study (MIDAS) Network Annual Meeting, Silver Springs, Maryland, November 2024

Reimagining the serocatalytic model for seasonal coronaviruses - 5 minute speed talk

Brook Lab, University of Chicago, Chicago, Illinois, November 2024

Host heterogeneity and SARS-CoV-2 dynamics: from structural inequity to immune history - 50-minute research talk

Epidemiology and Population Studies modeling seminar, NIH Fogarty International Center, Bethesda, Maryland, October 2024

Host heterogeneity and SARS-CoV-2 dynamics: from structural inequity to immune history - 50 minute research talk

Bohrer Memorial Student Workshop in Statistics, April 2024

Immune history influences SARS-CoV-2 booster impacts - 15-minute research talk and poster presentation

Microbe Mondays, April 2024

Host heterogeneity and SARS-CoV-2 dynamics: from structural inequity to immune history - 20-minute research talk

Annual Microbiology Research Conference, Monticello, Illinois, October 2023

30-minute research talk on host heterogeneity, social disparities, and impacts for SARS-CoV-2 control

Applied Math Graduate Seminar, University of Illinois at Urbana-Champaign, April 2022

25-minute research talk on social inequity and COVID-19 transmission MicroERA, University of Illinois at Urbana-Champaign, April 2022
Research speed talk on social inequity and COVID-19 transmission

POSTERS AT MEETINGS AND CONFERENCES

Ecology and Evolution of Infectious Diseases (EEID), Blacksburg, Virginia, June 2026

Same household, different choices: variation in health behaviors related to respiratory viruses in Illinois

Epidemics, San Diego, California, November 2025

Community connectivity and self-isolation as strategies to reduce the burden of respiratory diseases

American Society of Tropical Medicine and Hygiene (ASTMH) Annual Meeting, New Orleans, Louisiana, November 2024

Reimagining the serocatalytic model for seasonal coronaviruses

Institute for Disease Modeling (IDM) Annual Symposium, Seattle, Washington, September 2024

Reimagining the serocatalytic model for seasonal coronaviruses

Ecology and Evolution of Infectious Diseases (EEID), Stanford, California, June 2024

Disentangling the relative benefit of bivalent vs. monovalent boosters in the context of host immune history and socioeconomic status: a modeling study

MicroERA, University of Illinois at Urbana-Champaign, March 2024

Disentangling the seroconversion and seroreversion rates of seasonal coronaviruses using age-stratified seroprevalence data

Models of Infectious Disease Agent Study (MIDAS) Network Annual Meeting, Atlanta, Georgia, October 2023

Influence of immune history on booster impacts in low- and middle-income countries

ASTMH Annual Meeting, Chicago, Illinois, October 2023

Influence of immune history on booster impacts in low- and middle-income countries

MIDAS Network Annual Meeting, Bethesda, Maryland, September 2022

Funded by MIDAS to attend annual meeting and present on social inequity and COVID-19 transmission

Bohrer Memorial Student Workshop in Statistics, University of Illinois at Urbana-Champaign, April 2022

Poster presentation on social inequity and COVID-19 transmission

PUBLIC ENGAGEMENT

Science on Tap, Rigg's Brewery, November 2023

"Who gets sick and why: from COVID-19 to the common cold" - 1 hour discussion for a lay audience.

Genome Day, Champaign Public Library, November 2023

Two weekends of science exhibits and activities for a K-12 audience

SERVICE

Research Volunteer, The Ayah Project, October 2025 - present

Support research and education activities for the United States' first 501(c)(3) dedicated to supporting the LGBTQ+ Muslim community.

Campus Advisory Committee on the LGBTQ+ Communities at Cal, December 2025 - May 2026

Serve as one of two postdoctoral appointees. This is a standing committee for the UC Berkeley senior administration, advising on concerns of lesbian, gay, bisexual, transgender, intersex, and queer UC Berkeley employees and students. Member of the Health and Wellness subcommittee, helping to develop policy recommendations in response to student and employee needs.

MIDAS Trainee Committee Co-Chair, November 2024 - January 2026

MIDAS Trainee Committee, January 2024 - January 2026

Organize events for trainees in the Models of Infectious Disease Agent Study (MIDAS) Network

Disability Caucus, Graduate Employees Organization, Spring 2021 - Spring 2023

With Disability Caucus, which advocates for disabled students and workers on campus, developed a workshop on disability and the accommodations process aimed at teaching assistants, staff, and faculty. The workshop ran in multiple departments including as part of math department TA training.

Accommodation Working Group, August 2022 - October 2022

Contributed to early stages of a working group formed by the Department of Mathematics to develop recommendations of best practices for accommodating disabled students in the graduate program

Undergraduate Affairs Committee, Fall 2021 - Spring 2022

Served on the committee as the graduate representative. Helped make decisions about undergraduate curriculum and programs.

AD-HOC REVIEWER

BMC Infectious Diseases; Bulletin of the World Health Organization; Epidemics; Humanities and Social Sciences Communications; Journal of Health, Population, and Nutrition; Mathematical Biosciences; Proceedings of the Royal Society B; Scientific Reports

SELECTED COVERAGE

This Week in Virology, Episode 1035, August 2023: Poxvirus wars, warpspeed COVID-19 vaccine rollout

University of Illinois at Urbana-Champaign, School of MCB Communications, April 2024: Cheers to ‘Science on Tap:’ Graduate students demystify science topics at local brewery

Department of Statistics News, January 2023: Heat waves, stock prices, and COVID vaccines among student research projects presented at recent URES symposium